

VOZNESENS'K, Ukraine

WMO: 337770

Lat: 47.5825N

Lon: 31.3200E

Elev: 112

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 90-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	10.5	-1.5	5.1	6.9	3.3	6.5	12.2	17.2	32.5	15.1	33.8	4.0	330	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	92.6	68.9	89.1	67.5	86.3	66.6	71.1	85.3	69.7	83.7	68.4	81.9	4.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	99.7	75.6	65.3	94.0	74.5	63.7	88.8	73.8	35.1	85.1	33.8	83.7	32.7	81.9	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.7	9.9	-0.1	97.6	5.5	3.3	-4.0	100.0	-7.3	102.0	-10.4	103.9	-14.4	106.3		
(5)			-0.5	75.2	5.9	1.9	-4.8	76.5	-8.3	77.7	-11.6	78.8	-15.9	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	30.0	32.2	41.1	52.2	61.5	67.6	72.2	71.9	62.8	50.8	40.0	31.5		
	DBStd	17.12	9.77	9.39	7.82	6.79	6.07	5.53	5.10	6.15	7.43	8.71	9.56	8.21		
	HDD50	2457	619	499	291	54	3	0	0	0	4	97	316	574		
	HDD65	5682	1084	918	742	385	142	34	6	12	125	446	749	1039		
	CDD50	2919	0	1	14	121	359	528	690	680	388	121	17	0		
	CDD65	669	0	0	0	2	33	112	231	227	59	5	0	0		
	CDH74	7240	0	0	0	30	364	1035	2457	2669	657	28	0	0		
	CDH80	2730	0	0	0	3	63	312	927	1206	219	0	0	0		
	WSAvg	4.4	4.9	5.1	5.6	5.0	4.3	4.0	3.9	3.5	3.8	4.0	4.4	4.9		
	PrecAvg	19.3	1.3	1.1	1.1	1.2	2.0	3.1	2.3	1.5	1.6	1.5	1.2	1.5		
(15) Precipitation	PrecMax	33.1	3.3	3.1	2.4	2.8	4.4	8.1	6.3	3.5	4.8	5.3	2.8	3.3		
	PrecMin	11.9	0.4	0.2	0.1	0.2	0.1	0.8	0.4	0.3	0.0	0.1	0.1	0.1		
	PrecStd	4.7	0.9	0.7	0.6	0.7	1.3	1.8	1.5	1.1	1.3	1.1	0.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	58.9	68.4	77.2	84.6	89.7	94.9	96.8	92.8	77.6	62.3	52.7		
		MCWB	44.5	49.2	52.0	60.9	63.5	68.9	71.8	68.4	69.2	62.5	56.2	48.5		
	2%	DB	45.5	50.9	61.8	73.2	81.9	86.8	90.8	94.0	85.7	72.9	58.7	48.3		
		MCWB	42.1	45.1	49.7	55.9	63.0	67.3	68.4	68.5	65.2	60.2	54.0	44.5		
	5%	DB	42.8	46.6	57.1	69.1	78.2	84.0	88.0	90.6	81.3	67.7	55.7	44.4		
		MCWB	40.3	42.0	47.0	54.2	61.8	67.1	67.6	67.7	63.4	59.1	52.4	42.2		
	10%	DB	40.9	43.4	53.0	65.5	75.0	80.2	85.3	87.2	76.9	63.8	52.7	41.4		
		MCWB	38.6	40.0	44.5	52.8	59.7	65.6	66.8	66.3	62.1	56.6	49.7	39.5		
	0.4%	WB	45.2	49.8	53.8	62.4	67.5	72.5	74.2	72.4	70.3	66.2	57.9	49.1		
		MCDB	48.1	56.7	64.8	77.9	77.9	85.3	89.5	89.2	85.9	72.3	61.8	51.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.7	45.9	50.7	58.3	65.7	70.2	71.5	70.1	67.7	62.6	55.2	44.7		
		MCDB	44.6	50.5	60.3	68.7	76.3	81.7	85.5	87.2	82.0	69.8	57.8	47.6		
	5%	WB	40.6	42.6	48.1	56.0	64.1	68.7	70.0	68.8	65.7	60.1	52.8	42.2		
		MCDB	42.7	45.7	55.4	66.3	74.3	79.8	83.4	85.6	77.2	66.2	55.4	44.2		
	10%	WB	38.8	40.4	45.6	53.9	62.1	67.0	68.4	67.5	63.3	57.7	50.0	39.9		
		MCDB	40.6	43.4	51.6	63.5	71.5	77.2	81.0	83.7	73.5	62.7	52.4	41.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	12.0	15.2	18.6	20.9	19.7	22.2	23.6	21.0	17.3	11.2	8.7		
		MCDBR	11.3	16.8	23.0	25.4	25.8	25.0	26.1	29.4	27.5	23.2	14.6	12.4		
		MCWBR	9.0	10.7	13.3	11.9	10.3	9.2	8.8	9.6	11.1	13.2	10.1	9.7		
	5% WB	MCDBR	10.8	15.4	20.9	22.5	21.0	21.5	23.1	25.7	24.0	18.5	12.9	11.4		
		MCWBR	8.8	10.2	12.8	11.7	9.7	9.3	9.1	9.5	10.9	11.3	9.9	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.362	0.393	0.447	0.420	0.416	0.446	0.442	0.411	0.376	0.356	0.333		
	taud		2.259	2.208	2.213	2.093	2.255	2.290	2.215	2.219	2.271	2.351	2.332	2.315		
	Ebn at Noon		229	247	259	255	268	270	259	254	250	241	219	214		
	Edh at Noon		25	33	38	47	42	41	43	41	36	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		320	582	971	1455	1859	1989	1964	1761	1221	756	345	246		
	RadStd		54	91	113	129	158	128	117	99	159	92	59	52		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+1.24	N/A	N/A	+1.67	N/A	N/A	-198	-288	+216	+131

Nomenclature: See separate page